

Analysing qualitative data

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Research into behaviour in organisations often involves gathering and analysing qualitative data. Common sources of qualitative research data are:

- interviews
- responses to open-ended questions in questionnaires
- speeches or statements made by people we are studying
- documents that we analyse (company documents etc)

Coffey and Atkinson (1996, p. 36) say that when we analyse qualitative data, we are 'organizing the data into meaningful categories'. To do this, we need to either develop themes or categories that we can use for the analysis, or use categories that have been suggested by published theory.

This is a decision we need to take: whether to apply an existing theoretical framework to the qualitative data, or whether to develop a theory out of the data itself, or whether to use a mixture of these two approaches. For example, if we are analysing interview data as part of research into how people lead change, we might

- a) use existing frameworks of leadership theory to analyse the data (eg Transformational and Transactional Leadership Theory, Behavioural Styles Theory, Stage Theories of Change). This is a deductive approach, or
- b) look for themes in what our interviewees say about how they led others, or were led by them, and develop our theory from this data. This is an inductive approach (sometimes known as a **grounded analysis** - Easterby-Smith *et al.*, 2018), or
- c) use a mixture of the two approaches, looking for information that may be explained by the published theories, but also identifying and analysing any data that is not explained by published theory

Whichever of these approaches we choose, we will be looking for themes or categories in the qualitative data. If we choose approach a), these themes or categories will have been indicated by published theory; if we choose approach b), the 'grounded analysis' we will develop the themes ourselves, by analysing the data; if we choose approach c), we will be taking some themes from publications and developing some themes ourselves. Note that if we take approach b), we will need at some point to compare what we have found with published theories on leadership - we do not ignore what is already known about the area we research.

Suppose we decide to use Bass's framework of transformational and transactional leadership to analyse our interviews; our categories for analysis would certainly include the different dimensions of that framework, ie:

- idealised influence
- inspirational motivation
- intellectual stimulation
- individualised consideration

- contingent reward
- management by exception (active)
- management by exception (passive)

- laissez-faire leadership

We would analyse the data for evidence of each of these dimensions.

The research may provide new insights into the framework as we use it - for example, we might find that there are details emerging of sub-dimensions (different types of inspirational leadership, for example) or indications about situations where each dimension was commonly used, or signs of particular connections between the dimensions.

We might also develop other categories to use in the analysis - for example, to assess the extent to which the leader's actions had a positive impact on others, or achieved successful results for the team or the company.

You can see from this, that it would be useful to know, in advance of carrying out the interviews, how you plan to analyse them.

Where we are working with small numbers of cases - for example, if our research data is drawn from a small number of semi-structured interviews - we are most likely to use some form of **thematic analysis**. Through this, we may explore how our interviewees acted or thought or spoke about particular events. Semi-structured interviews provide a good opportunity to gather rich, detailed information about what has happened in particular situations.

For example, if we are carrying out detailed interviews with a small number of people, to research how people lead change, and we are using the Bass framework of Transformational Leadership, we might find examples of different ways in which this small number of people go about the 'intellectual stimulation' activity, and examples of how this is linked with 'individualised consideration' behaviours. Or we might find new evidence about one of the dimensions - or find sets of activities that appear to have been very important, but which are not included in Bass's framework.

Because these findings are based on research with only a small number of people, we will not be able to claim that they apply to all leaders of change (not even to those leaders of change in similar positions to our interviewees). But we can show that they apply to our interviewees - and that therefore it is *possible* they might apply more widely - and note that more research, with larger numbers of people, would be needed to investigate this.

With small numbers of interviewees, quantitative analysis is not usually very important. If we interview eight people, and all eight show certain specific behaviours, this is interesting; but behaviours that are shown only by one interviewee may also be interesting and significant, if they throw new light on an aspect of leading change.

Where we are analysing qualitative data from larger numbers of people (for example, answers to open-ended questions in a survey) then quantitative measures become more relevant. We may then count the number of times a particular theme is present in an answer. This approach to analysing qualitative data is called **quantitative content analysis** (see Easterby-Smith *et al.*, 2018). Counting occurrences may also be relevant for smaller numbers: it may be interesting that all of our interviewees show a particular behaviour - even if there are only eight of them.

The process of analysing qualitative data begins with familiarisation - ie reading the transcripts, listening to the recordings, immersing yourself in the data. Next comes some work on 'coding' - interpreting the data you have gathered. If you are using a set of categories from a published model or framework (such as the Bass framework, above), this means identifying these categories in the

data. If you are carrying out a grounded analysis, this means developing your own codes and categories from the data. A grounded analysis is often an iterative process, resulting in a large number of codes at first, which can then be combined into a smaller range of larger themes or categories. As you develop your themes, you may wish to re-code earlier material, to check on your analysis of it (and to test the changes to the codes). The analysis may then enable you to develop ideas about the details of the themes and linkages between themes.

The stages can be described as:

1. Familiarizing yourself with your data	Repeated listening to recording, reading and re-reading notes or transcripts, noting first ideas.
2. Generating initial codes	Coding interesting features of the data systematically, collating data relevant to each code.
3. Searching for themes	Bringing together codes into potential themes.
4. Reviewing themes	Checking if the themes work a) in relation to the coded extracts and b) the whole data set.
5. Defining and naming themes	Ongoing analysis to refine the specifics of each theme, generating clear definitions and names for each theme, and reviewing the overall story produced by the analysis.
6. Producing the report	Selection of good extracts as examples, final analysis of extracts, relating the analysis back to the research question and literature, and producing a scholarly report of the analysis.

(Braun and Clarke, 2006, p. 87)

None of this is mechanical and entirely straightforward - it all requires careful thought and interpretation, and it is likely to involve some experimentation. When you write up your research, you need to explain the process you went through in analysing the data.

Coding

Coding is a key part of analysing qualitative data. Codes pick out parts of the data that are relevant to your research. By coding the data 'you are *organising* your data into meaningful groups' (Braun and Clarke, 2006, p. 88).

For example, in one research project, senior managers interviewed about how they achieved business transformations in their companies said:

'We then went through a restructure where we created what we called a 'core team' [...] There was just much more ownership and accountability in the programme from people that had the power to actually make decisions.' (P3-I6 = Project 3, Interviewee 6)

'... at some points we expanded to a team of between 15-30 people and then we would contract - but the 'core team' was that team of five or six people with the project manager and myself.' (P6-I7)

'[...] the decision-making forum was five or six people' (P7-I9)

'I don't believe there is one individual that [...] if I think of the multi-disciplined team we had - there were various people that - you know people have blind spots - so I think it's about that almost that entire team synergy - and not just about one individual leading the change.'
(P8-I10)

(Baxter, 2019)

These eventually formed a theme within the research, of the transformations being led by a core team of 5-6 people. The quotes could be initially coded in different ways (such as, gained ownership and accountability through team working; leadership team size expanded and contracted; key decisions taken by a team of 5-6; need for a team to lead, not an individual). As it became apparent that the use of a 'core team' was a common phenomenon that was perceived to be an important part of the leadership of the transformations, the codes were brought together in a theme (see Braun and Clarke, 2013).

One approach to coding is to work through your data in detail, being careful to focus on its relevance to what you are trying to find out in your research. If you are carrying out inductive coding, in your initial codes generalise just a little, stay quite close to the data.

For example, in the quote above: 'We then went through a restructure where we created what we called a 'core team' [...] There was just much more ownership and accountability in the programme from people that had the power to actually make decisions.'

A possible code is: 'gained ownership and accountability through team working'. This strips away some of the detail of the original quotation, but still retains some key content.

Make sure you can locate the quotations associated with the codes: some people use colour highlighters, others notes in the margins of transcripts, others use tables in Word, or Excel spreadsheets, or programmes such as nVivo.

Compare codes and quotations as you go, noting similarities and differences.

This approach will generate a large number of codes, which you will then need to bring together in themes.

Presenting findings

When you present your findings, it is good practice to give examples of data that you have placed in each of your themes. For example, if your interviewees have made a number of statements that you have interpreted as being about leaders showing individualised consideration, then quote some examples in the text, or in a table in your text (as in Table 1, below). This is a way of demonstrating your analysis to the reader, and it enables the reader to assess how you have linked your data with the themes you claim it shows.

You can see in the example below that, rather than simply coding against individualised consideration, more detail has been retained, enabling the analysis to demonstrate different types of individualised consideration.

Table 1: Examples of individualised concern

Individualised concern: personal	'My manager said she thought I might need more time to recover, as she knew it had been a difficult time for me...' (Int B2) 'Of course, I knew he was under pressure at home, so I suggested he get some help from D....' (Int C5) 'He nearly always stops by the office on his way in each morning and we have a chat - often about nothing to do with work...' (Int D1)
Individualised concern: career development	'He took a lot of time talking with me about where I saw my career going in the next five years....and then helped me to plan the first steps...' (Int A3) 'I try to link what they are doing in the short term with what they want to develop in the longer term...and talk with each of them about that...' (Int A4) 'She really helped me to think through the difficult issues that were involved in that project, just by asking questions, acting as a sounding board, really...helping me to work it out for myself...' (Int D2)

A note on terminology

There is a wealth of published advice by experienced researchers on the process of making sense of qualitative data. There are some similarities and some differences in the approaches they advocate. Most, if not all, researchers suggest a process of developing (or applying) a first set of codes to the data, and then refining these codes – often bringing together these initial codes in combinations that express their relationship to one another. However, the terms that are used for different stages of this process of analysis often differ from writer to writer. This paper has talked of ‘codes’, which can then be combined into a smaller number of ‘themes’ or ‘categories’.

Braun and Clarke (2006) talk of combining ‘codes’ into ‘themes’, as do Robson and McCartan (2016). In a more recent paper, Braun and Clarke (2020: 14) define themes in their approach to thematic analysis as ‘patterns of shared meaning, united by a central concept or idea’.

Easterby-Smith *et al.* (2018) indicate that ‘codes’ may be combined into a smaller number of ‘categories’, which will in turn be combined into an even smaller number of ‘themes’. Marshall and Rossman (2011) appear to use ‘codes’ and ‘themes’ interchangeably. Coffey and Atkinson (1996) talk of ‘more general’ and ‘more specific’ codes, and consider the more general codes to be the same as ‘categories’; and they appear to use ‘categories’ and ‘themes’ interchangeably. Richards (2015) talks of ‘coding’ as a process of ‘creating categories’, so that his ‘categories’ are what other writers call ‘codes’. Charmaz (2006) explains that a process followed in grounded theory research will be to develop ‘initial codes’, to then select from these a smaller number of ‘focused codes’, and from these develop a smaller number of ‘theoretical codes’, which explain how the focused codes may relate to each other.

So there is no clear agreement about terminology. In your work on analysing qualitative data, decide what you will call your units of analysis, preferably referencing that you are following the terms used by a published source, and then be consistent.

References and further reading

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- C Robson and K McCartan 2016 *Real World Research*, Wiley, see ch 18 'The analysis and interpretation of qualitative data' pp. 459-486